22AIE111 OBJECT ORIENTED PROGRAMMING IN JAVA
S2 B.Tech CSE-AI
Department of Computer Science and Engineering
Amrita School of Computing, Amritapuri Campus
CASE STUDY
PHASE – 2 (Design)
Architectural Modelling (CO2)

Group No : **12**

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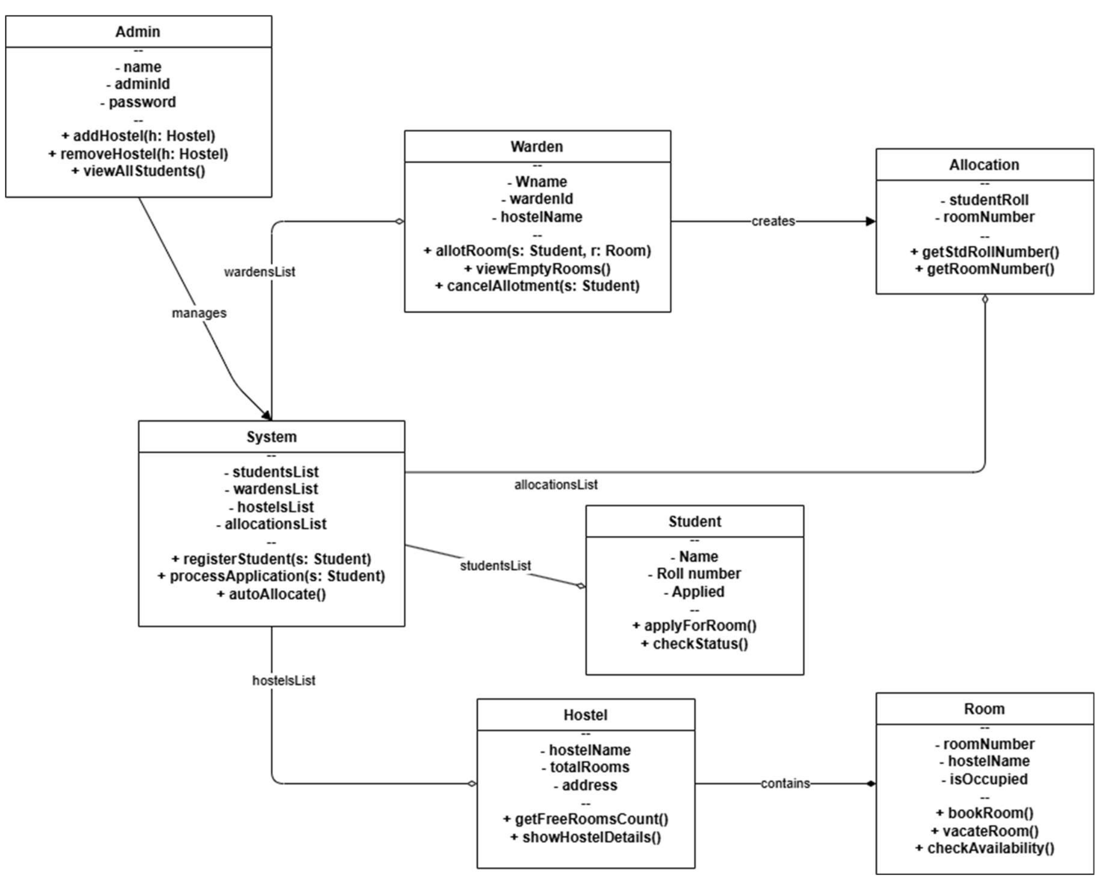
1. Problem Title : **Hostel Management System**

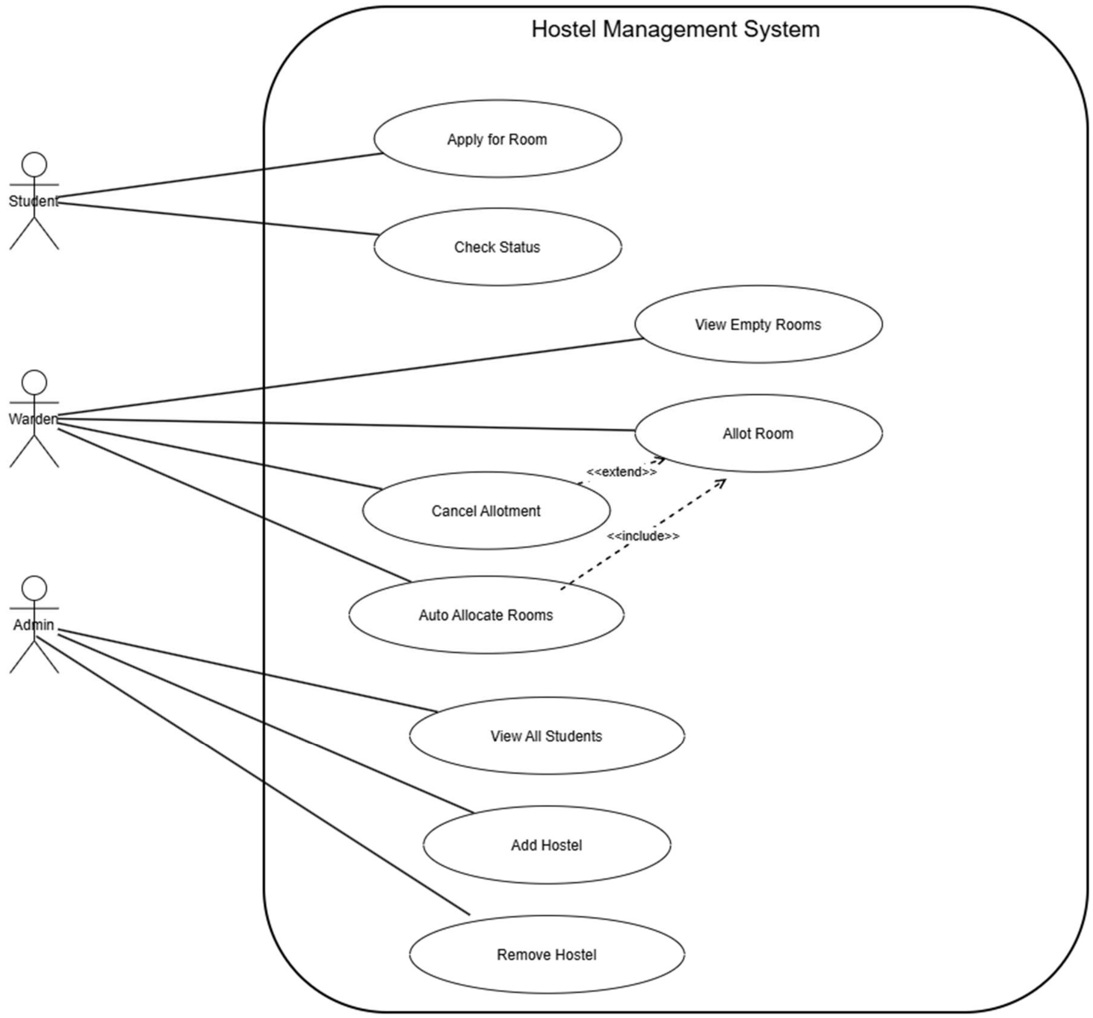
STRUCTURAL DESIGN

2. Class diagram (*draw class diagram that shows different classes and their relationships. Use UML notations*)



<u>BEHAVIUORAL DESIGN</u>

3. Use-case diagram
   a. List of actors
      1. **Student:** The primary user seeking accommodation.
      2. **Warden:** The managerial user responsible for day-to-day room assignments and hostel operations.
      3. **Admin:** The high-level configuration user who sets up the digital infrastructure.
   b. List of use-cases
      1) Register Student: System records student credentials and details for authentication and future processing.
      2) Apply for Room: Authenticated student submits a room request, setting their status to "pending."
      3) Check Application Status: Student queries the system to view the current progress (pending, approved, or assigned) of their request.
      4) Process Application: System validates student data and queues it for warden review.
      5) Allot Room: Warden assigns a vacant room to an approved student, creating a permanent allocation record.
      6) Cancel Allotment: Warden revokes an existing room assignment, deleting the mapping and freeing the room.
      7) View Empty Rooms: Warden retrieves a list of all currently unoccupied rooms within their hostel.
      8) Auto Allocate: System automatically maps all pending applications to available rooms in a batch process.
      9) Add Hostel: Admin instantiates and saves a new hostel building record into the system.
      10) Remove Hostel: Admin deletes an existing hostel record and all its associated data.
      11) View All Students: Admin generates a comprehensive report of every student registered in the system.
      12) View Allocated Students: Warden views a list of students successfully assigned rooms within their specific hostel.
      13) Vacate Room: System removes the allocation link and marks a room as vacant upon student checkout.
   c. Use case diagram (use UML notations)

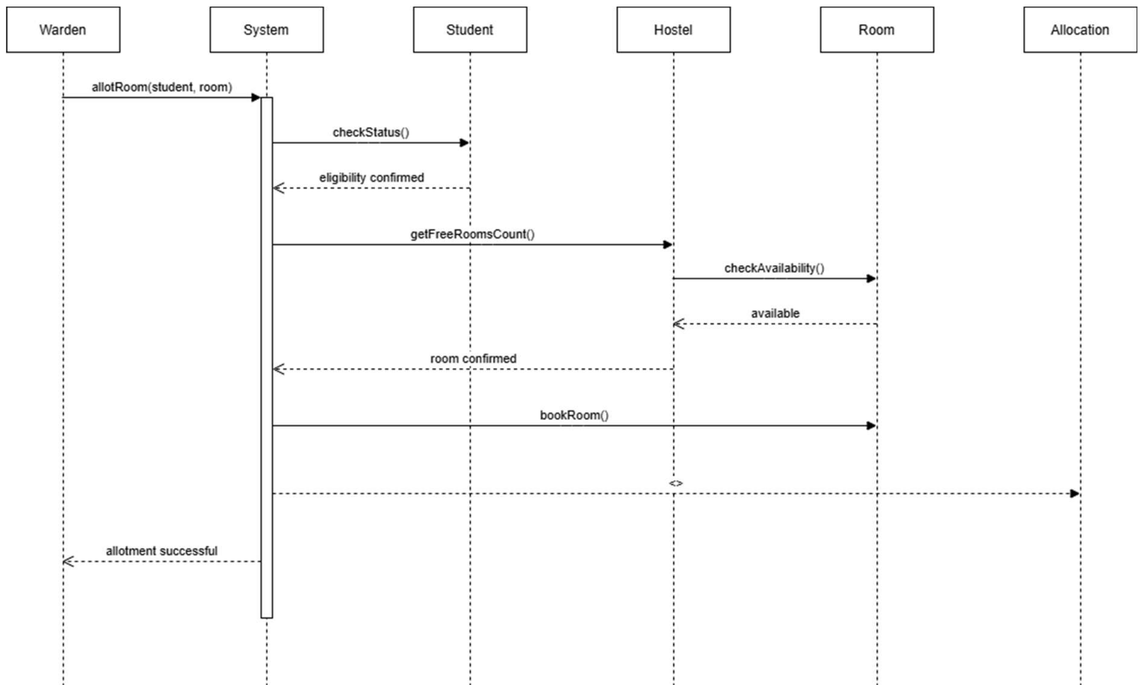


1. Sequence diagram (use UML notations)

   a.

   Scenario 1: Room Allocation Process

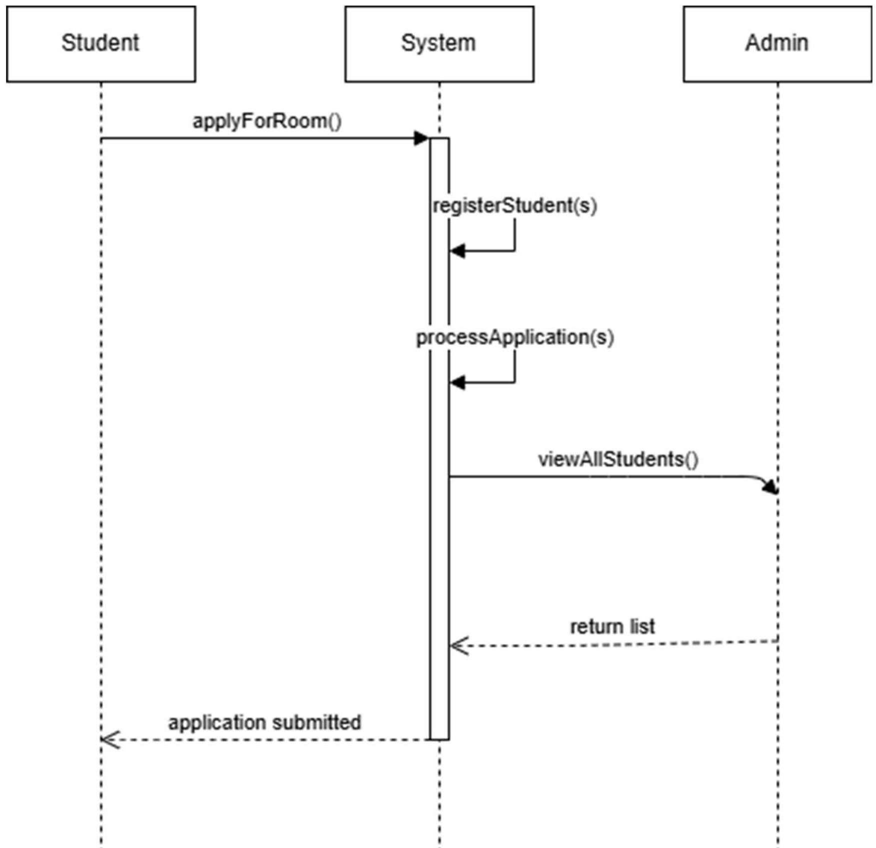
   Scenario Description: The Warden initiates a room allotment for a student. The System validates the student's status, checks the Hostel for available capacity, confirms the specific Room is vacant, and finally creates a new Allocation record.



b.

Scenario 2: Student Application Submission

Scenario Description: A Student applies for a room through the System. The System registers the student into the database, processes the application logic, and allows the Admin to view the updated student list for oversight


Student
System
Admin
applyForRoom()
registerStudent(s)
processApplication(s)
viewAllStudents()
return list
application submitted